

WASHINGTON EV MARKET OVERVIEW

Executive Tableau Analytics Case Study

Independent portfolio project using public vehicle-registration data

A decision-oriented market analysis designed to help leaders understand geographic concentration, manufacturer performance, fleet composition, and data-quality limitations across Washington State.

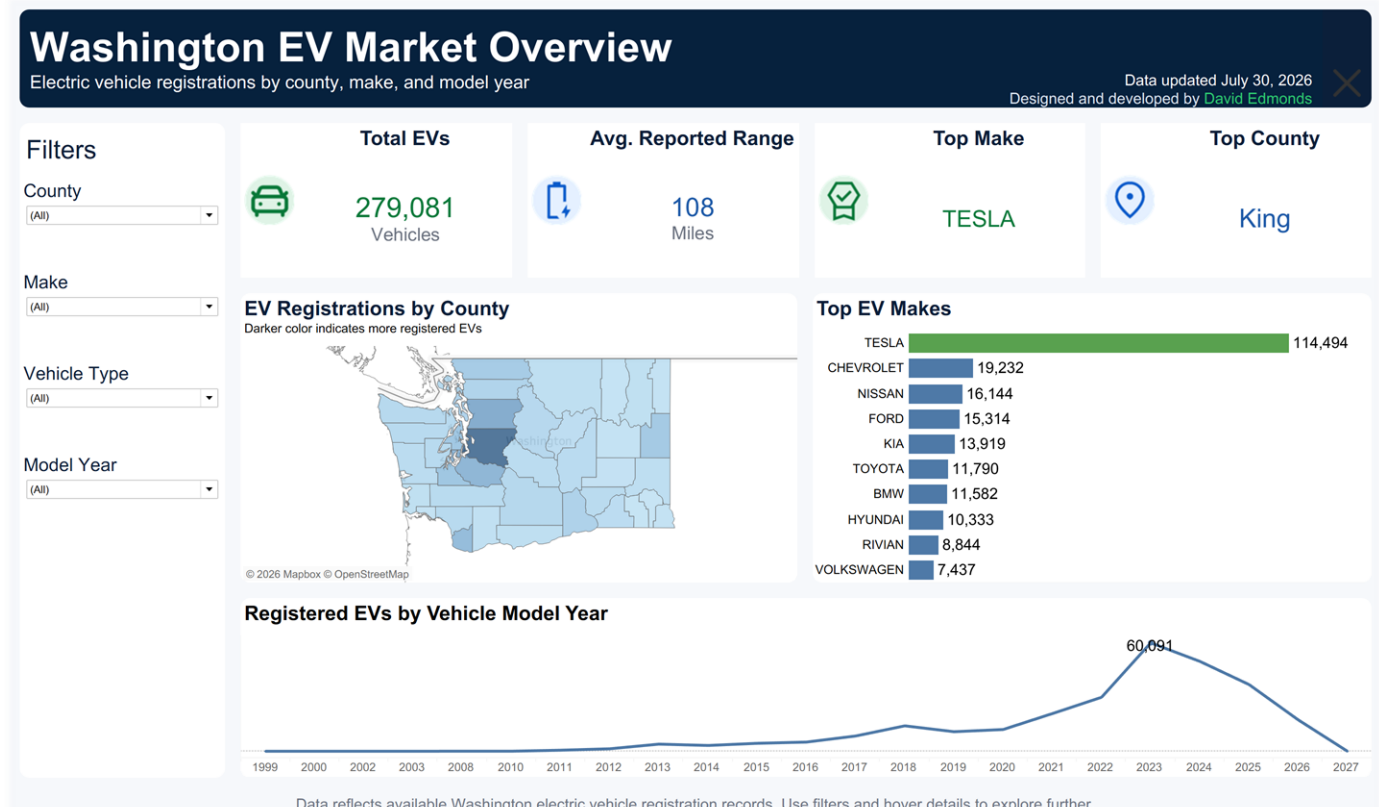


Figure 1. Interactive executive dashboard with collapsible filters, dynamic KPIs, geographic analysis, manufacturer ranking, and model-year exploration.

ROLE	PRIMARY TOOL	DATA EXTRACT	FOCUS
Senior Analytics Consultant	Tableau	July 2026	Market & operational insight

Executive Brief

Bottom line

The raw vehicle-level dataset contained valuable market information but was not structured for rapid decision-making. I designed a self-service Tableau product that converts the data into governed KPIs, dynamic rankings, geographic concentration analysis, and model-year insights while clearly separating valid metrics from incomplete source fields.

BUSINESS NEED	ANALYTICAL RESPONSE	DELIVERED VALUE
Leaders needed a concise view of where EVs are concentrated, which manufacturers lead, and how the registered fleet is distributed.	I defined the questions, validated the data, governed KPI logic, designed the visual hierarchy, and built interactive Tableau actions.	The final dashboard supports statewide monitoring and rapid drill-down by county, make, vehicle type, and model year.

Engagement Context

Washington’s electric vehicle population dataset contains detailed records by county, manufacturer, vehicle type, model, model year, and reported electric range. The information can support infrastructure planning, market assessment, fleet strategy, and public-policy analysis, but the raw file does not provide a clear leadership view.

I approached the project as a consulting engagement rather than a chart-building exercise: first define the decisions the analysis should support, then establish trustworthy metrics, and finally design an interface that allows users to move from statewide context to a specific segment without losing orientation.

Core Questions

- Where are registered electric vehicles concentrated across Washington?
- Which manufacturers hold the strongest position overall and within selected segments?
- How is the registered fleet distributed across vehicle model years?
- How do results change when a user filters by county, make, vehicle type, or model year?
- Which source-data limitations could materially distort the interpretation of key metrics?

Analytical Approach

I structured the work across four consulting phases: diagnostic review, metric governance, solution design, and validation. This ensured the dashboard was built around decision quality rather than visual preference alone.

1 DIAGNOSE	Profile the fields, identify incomplete values, test record counts, and determine which dimensions could support reliable segmentation.
2 GOVERN	Define each KPI, document inclusion and exclusion rules, and prevent a data-quality filter from unintentionally changing unrelated measures.
3 DESIGN	Build an executive hierarchy: summary KPIs first, comparative analysis second, and detail through filters, selections, and tooltips.
4 VALIDATE	Test calculations and actions under multiple combinations, confirm Top-N behavior, and review how selected marks change the supporting KPIs.

Data Preparation and Metric Governance

The most important analytical issue was not the visual design; it was ensuring that missing or placeholder values did not contaminate the dashboard's logic.

Reported electric range

The source contained null and zero values for electric range. Including those records in the average materially understated the metric. I excluded null and zero values only from the Avg. Reported Range calculation while preserving all valid vehicle records in registration totals, county analysis, manufacturer rankings, and model-year analysis.

Model-year interpretation

Model year describes the vehicle's model year, not the date it was registered. I used the title "Registered EVs by Vehicle Model Year" to reduce the risk that viewers would interpret the chart as a historical registration trend.

Geographic color scaling

King County has a substantially larger EV population than most counties. A standard linear color scale compressed the rest of the state into nearly identical shades. I used a transformed color measure to improve visual separation while preserving the true vehicle totals in the tooltip. The map therefore communicates concentration without changing the underlying values.

Solution Design

The dashboard was designed for fast executive scanning first and deeper exploration second. The layout places four summary indicators at the top, market and geographic comparisons in the center, and the model-year distribution across the full width at the bottom.

COMPONENT	BUSINESS PURPOSE	DESIGN DECISION
KPI cards	Establish the current statewide baseline and leading dimensions.	Dynamic totals, average range, top make, and top county update with the user's selections.
County map	Show where EV registrations are concentrated.	Transformed shading improves separation; exact totals remain available on hover.
Top manufacturers	Compare competitive position and market concentration.	Dynamic Top-N logic recalculates within the selected context; only the leader is highlighted.
Model-year profile	Show the age and composition of the registered fleet.	The peak is labeled by default; a selected year receives the label during interaction.
Filter panel	Support self-service analysis without permanently reducing chart space.	A show/hide control reveals county, make, vehicle type, and model-year filters only when needed.

Advanced Tableau Functionality

- Context-aware Top-N manufacturer rankings that recalculate after filters are applied.
- Dynamic top-make and top-county cards that return the statewide leader by default and display the selected mark during exploration.
- A model-year set action that labels the selected year and returns to the peak year when the selection is cleared.
- A collapsible filter panel that preserves dashboard space while supporting detailed segmentation.
- Scoped filters that prevent data-quality corrections from changing unrelated totals.
- Interactive tooltips that expose exact values without cluttering the primary view.

Quality Assurance

I tested the dashboard across multiple selections and filter combinations to confirm that summary KPIs, geographic shading, manufacturer rankings, and model-year labels remained logically consistent. Particular attention was given to filter order, action behavior, and the difference between statewide default views and user-selected segments.

Key Findings

Geographic concentration

EV registrations are heavily concentrated in King County, with additional volume clustered in other higher-population counties. Statewide totals therefore mask substantial geographic differences in market maturity and likely infrastructure demand.

Manufacturer dominance

Tesla holds a clear lead in the current vehicle population. The remaining market is distributed across a wider group of manufacturers, indicating a dominant leader followed by a fragmented competitive field.

Newer fleet composition

The registered fleet is concentrated in more recent model years, with the highest volume in the early-to-mid 2020s. The newest model years should be interpreted cautiously because data may be incomplete as records continue to enter the system.

Data-quality sensitivity

The reported electric-range average changes materially when missing and placeholder values are handled correctly. This confirms that metric governance is necessary before source fields are presented to decision-makers.

DECISION IMPLICATIONS

Strategic Recommendations

Prioritize high-volume counties for near-term planning: Use the county concentration analysis to guide charging infrastructure, utility planning, fleet services, and outreach where the immediate EV population is largest.

Add a population-adjusted adoption rate: Pair vehicle totals with population or household data to distinguish counties with the largest absolute markets from counties with unusually high adoption relative to their size.

Separate BEV and PHEV demand: Analyze battery-electric and plug-in hybrid vehicles independently because their charging behavior and infrastructure requirements differ.

Add time-based registration history: Model year shows fleet composition but not the timing of adoption. Historical extracts or registration dates would support a true growth and forecasting view.

Integrate charging-station supply: Compare registered vehicles with public charging availability to identify potential infrastructure gaps and underserved markets.

Outcome and Next Phase

The completed solution converts a detailed public dataset into an executive-ready analytical product. It establishes a statewide baseline, supports interactive segmentation, communicates data limitations, and demonstrates how governed calculations can be integrated into a clean self-service experience.

Delivered outcome

A portfolio-quality Tableau dashboard that demonstrates senior-level ownership across problem framing, data validation, KPI governance, dashboard engineering, interaction design, quality assurance, and strategic interpretation.

Current Limitations

- The dataset reflects available vehicle-registration records rather than a complete historical adoption timeline.
- Vehicle model year should not be interpreted as registration year.
- Reported electric range is incomplete for some records.
- County totals are not adjusted for population or household count.
- Vehicle registrations alone do not measure charging availability or grid capacity.
- The newest model years may be incomplete.

Recommended Next Phase

- Add population-normalized adoption metrics by county.
- Incorporate public charging-station locations and charger counts.
- Develop historical snapshots to measure actual registration growth over time.
- Add a market-share view by make and vehicle type.
- Publish the dashboard with a documented data dictionary and refresh process.

Capabilities Demonstrated

Executive dashboard design	KPI and metric governance	Data-quality validation
Tableau calculations and actions	Geographic market analysis	Dynamic Top-N reporting
Stakeholder-focused communication	Analytical storytelling	Strategic recommendations

Consulting takeaway

My approach to analytics consulting is to define the decision that needs to be supported, validate the information before it reaches leadership, build metrics that users can trust, and present the result through a clear, interactive reporting experience.

PROJECT REFERENCES

Data: Washington State Electric Vehicle Population Data | Tools: Tableau, GitHub, CSV data preparation